



Robotic actuators powered by steam

Researchers at the University of Coimbra (Portugal) have developed a modular soft actuator powered by steam from boiling water. It uses a 24V heater to generate pressure rapidly (100kPa/sec), delivering over 50N of force. This



An untethered soft robot crawls along a tree trunk in an unstructured environment. It is powered by four liquid-gas phase transition electric soft actuators that use water as the working fluid (Credit: Diogo Fonseca)

simple, low-cost design avoids pumps and harmful materials. Using safe, widely available water, the actuators are fast, durable, and adaptable for various motions. Demonstrations include a gripping hand, a soft claw, and Bixo, a tree-climbing robot.

Printable aluminium alloys for automobiles and aircraft

Researchers from CMU (Pittsburgh, PA) and the Massachusetts Institute of Technology (Cambridge, MA) have developed a new printable aluminium alloy system using Al-Ni-Er-Zr-Y, designed through simulations and machine learning. This alloy retains up to 95% of the strength of top-performing alloys while reducing costs by 15%, with a room-temperature version costing just 20% of the price. Optimised for additive manufacturing, it offers superior strength under rapid cooling. The innovation enables the development of stronger, cheaper, and more sustainable materials for automotive and aerospace applications.

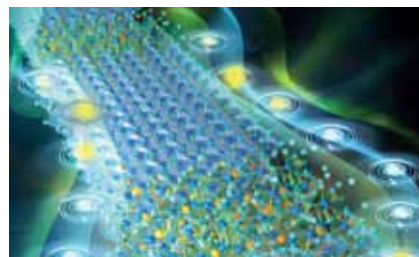
High-resolution microscope for heat flow measurement

Scientists from the Technical University of Denmark (DTU), Technion (Haifa, Israel) and the University of Antwerp (Belgium) have developed a high-resolution microscope that tracks directional heat flow in thermoelectric materials. This non-destructive tool, based on the CAPRES microRSP platform, enables accurate nanoscale heat measurements without requiring special sample preparation. It improves

understanding of heat transport, helping to optimise devices like solar panels and batteries, and enhancing thermoelectric efficiency, which is crucial for developing faster electronics and supporting the green energy transition.

Material design for pure-red perovskite LEDs

A research team from the University of Science and Technology of China (USTC), affiliated with the Chinese Academy of Sciences, has developed a 3D intragrain heterostructure for pure-red perovskite LEDs (PeLEDs), featuring



Material design for red perovskite LED (Credit: www.perovskite-info.com)

ing embedded narrow-bandgap emitters and PTLA for lattice stabilisation. This design confines charge carriers, preventing hole leakage and enabling efficient transfer. Using

EETA spectroscopy, they pinpointed leakage issues. The innovation delivers record-breaking performance, 24.2% peak EQE, 24,600cd/m² luminance, 10.5% EQE at 90% brightness, and 127-hour half-lifetime—solving brightness-efficiency trade-offs in PeLED technology.

Sensors that last in blood

Researchers at Stanford University (California) have developed the SENSBIT (stable electrochemical nanostructured sensor for blood in situ tracking) system, a modular biosensor implanted directly into blood vessels, to overcome



A microscopic view of the large intestine shows the microvilli lining that inspired the protective structure of the SENSBIT biosensor (Credit: <https://news.stanford.edu>)

the instability of traditional biosensors in the bloodstream. Inspired by gut defences, it features a 3D nanoporous gold surface and a mucosa-like coating, enhancing its durability. SENSBIT remained functional for a week in live rats,

outperforming previous devices. This innovation enables early detection of infections and continuous, real-time monitoring, advancing personalised medicine and long-term molecular tracking.